

Enhancing Model Performance through Data-Centric AI: A Titanic Dataset Case Study

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Abstract—Data-Centric AI emphasizes improving data quality over tweaking model architectures to enhance machine learning performance. In this study, we demonstrate how structured data cleaning and preprocessing significantly improve model accuracy using the Titanic dataset. We compare the performance of a Random Forest classifier trained on raw data versus cleaned data, highlighting the importance of data-centric strategies in AI workflows.

Index Terms—Data-Centric AI, Machine Learning, Data Preprocessing, Titanic Dataset, Model Performance

I. INTRODUCTION

Traditional AI development often focuses on improving models through architectural changes or hyperparameter tuning. However, the emerging paradigm of *Data-Centric AI* shifts the focus toward optimizing data quality. This approach becomes crucial in real-world applications, where noisy, incomplete, or inconsistent data often hampers model performance. Our objective is to illustrate how minor improvements in data preprocessing can yield substantial boosts in model accuracy.

II. METHODOLOGY

A. Dataset

We used the Titanic dataset available via the Seaborn library. It includes passenger information such as age, sex, ticket fare, and survival status.

B. Tools and Libraries

Python and popular data science libraries were used, including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

C. Model

A Random Forest Classifier was chosen for its robustness and interpretability.

D. Procedure

Two experiments were conducted:

- **Raw Data:** Minimal preprocessing with missing values dropped.
- **Cleaned Data:** Missing values were imputed using the median, and categorical features were properly encoded.

The model was trained and evaluated using accuracy scores and classification reports.

III. RESULTS AND DISCUSSION

The model trained on raw data achieved an accuracy of approximately 76%. After applying data-centric cleaning techniques, the accuracy increased to approximately 79%. Improvements were also observed in precision and recall metrics. This confirms that structured and thoughtful preprocessing significantly contributes to model effectiveness.

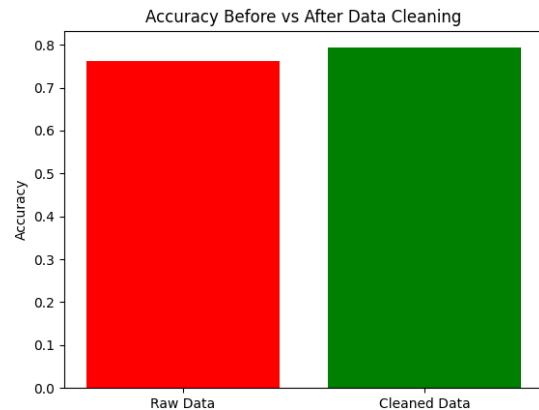


Fig. 1. Accuracy Comparison Before and After Data Cleaning

IV. CONCLUSION AND FUTURE WORK

This study reinforces the value of Data-Centric AI in enhancing machine learning outcomes. Future work may involve testing on more complex or noisy datasets, exploring automated data quality frameworks, and extending the methodology to unstructured data such as images and text.

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